AI OPEN
DATA CENTER



Flexibility the Grid Can Count On

Turning Curtailable Load Into Interconnection Capacity

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The bottleneck isn't megawatts. It's the queue.

AI's binding constraint has moved from generation to interconnection

The U.S. can generate power. It cannot connect new load fast enough to use it.

Grids are sized for the worst hour of the worst day, then run far below that limit almost all the time.

That unused headroom is real capacity, if new load will yield during the few hours power is scarce.

Flexibility the grid can count on turns a firm-power request into an interconnection strategy.

The grid already has room for flexible load

Norris et al., Duke Nicholas Institute, "Rethinking Load Growth," February 2025. 22 balancing authorities, 95% of U.S. peak load. 76 GW at 0.25% curtailment ≈ 10% of U.S. peak demand.

76

GW at 0.25% annual curtailment
(85 hours per year)

98

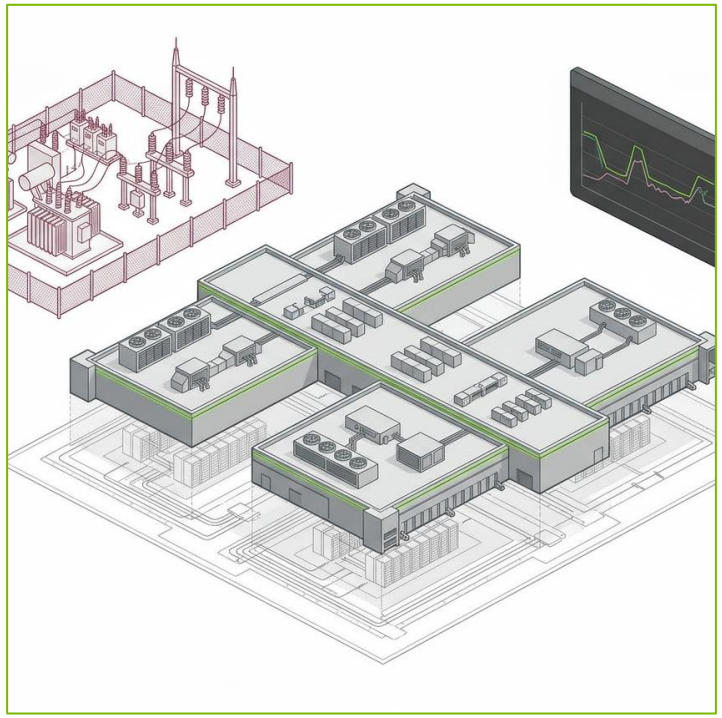
GW at 0.5% annual curtailment
(177 hours per year)

126

GW at 1.0% annual curtailment
(366 hours per year)

AI Factory Flex

Secure, dynamic power management between the grid and the AI Factory



Key Benefits

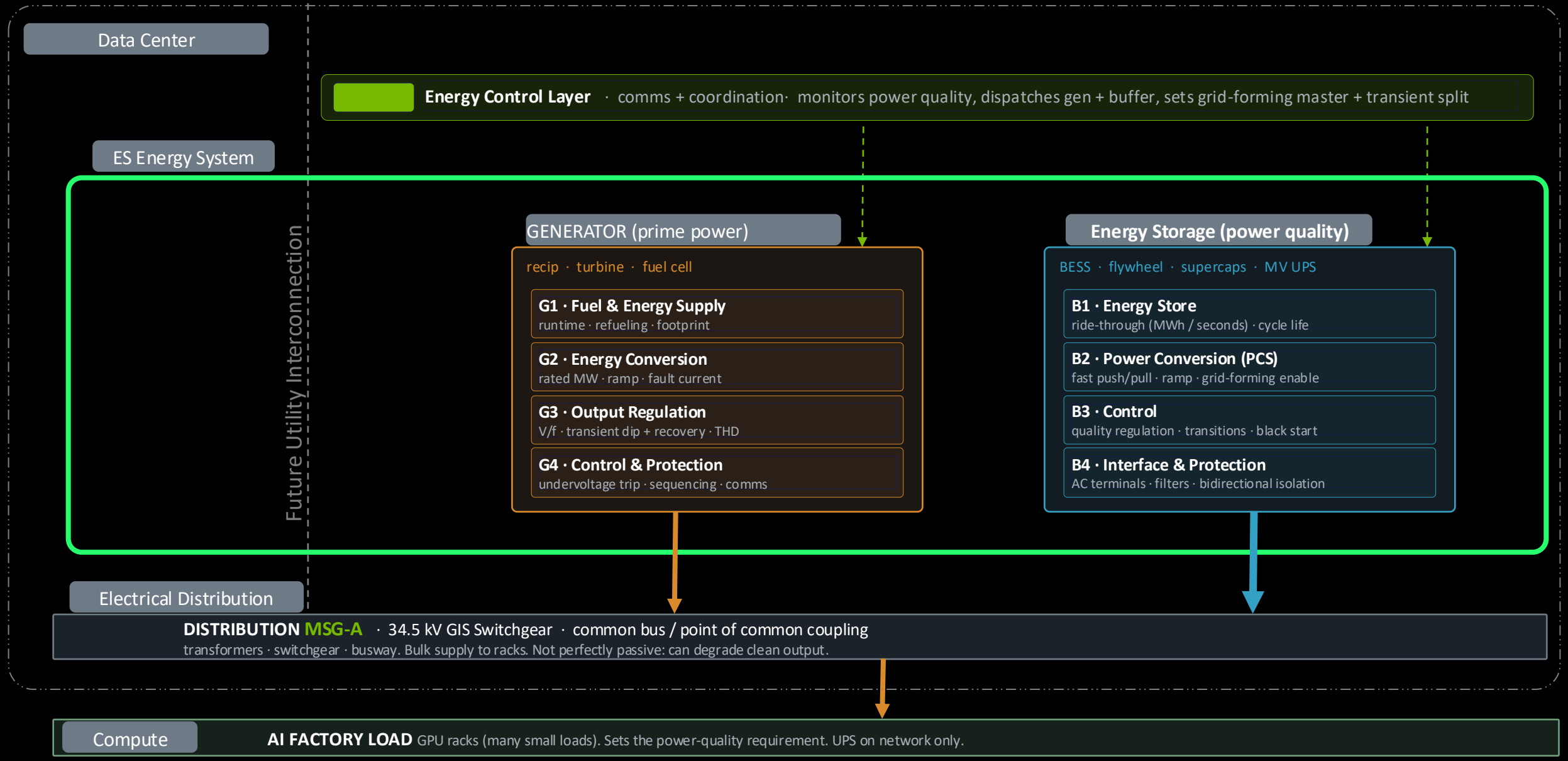
Balances AI load with the grid
Dynamically adjusts AI workloads to match real-time grid conditions.

Enables renewable and hybrid power
Orchestrates power from the utility, on-site renewables, and storage.

Improves reliability and economics
Reduces curtailment risk, avoids peak demand penalties, and keeps GPUs productive.

The Bridging BTM Power: Generator + Buffer System Architecture

What it takes to plug a data center into on-site power. Energy flows left to right; DSX coordinates; the buffer holds power quality.



Flexibility only counts if it doesn't break the SLA

Why blunt curtailment fails the buyer, and what the grid needs

An AI factory sells uptime. Forced blackouts break the contract and the customer.

0.5% curtailment sounds trivial, but 177 hours of hard downtime a year violates typical availability terms.

The grid needs load it can lean on. The buyer needs performance it can promise. Both are right.

The answer is not dumb curtailment. It is graceful, bounded, verifiable flexibility.



Shift the Work

Move or delay non-urgent training and batch jobs across time and sites. Protected workloads keep running.



Lean on Storage & Onsite Power

BESS and co-located generation absorb the swing as bridge power, then support the grid instead of sitting idle.



Trim Power & Cooling

Tune GPU power caps and cooling to a set target in seconds, never below the agreed floor.

Energy Storage System Role & Requirements

Sai Somayajula



Energy Storage Role

- Facility Level Buffer
  - Meet POI Requirements
- Abnormal Event Response
  - Voltage ride through, voltage & frequency stabilization
- Bridge Power
  - Source transition & black start capability
- Grid Dispatchability
  - Grid capacity relief and flexibility

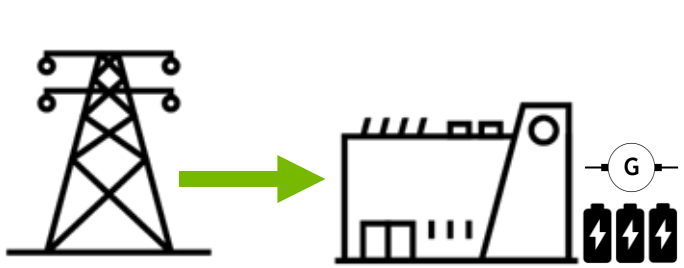
Energy Storage Sizing & Controls

- Sizing Criteria
  - Frequency targets, ramp rates
  - Grid / Gen requirements, Facility Load Profile
- Controls
  - Grid forming mode
  - Latency & Interface requirements
- Battery Chemistry
  - C-rate, State of Charge, Degradation

Energy Storage Functions per Power Option

Energy Storage Role		Source of Power Options	
Function	Parameter	1 - All Grid Connected 2b - Flexible Grid Asset	2a - Bridge-to-Grid 3 -All Islanded
Facility Level Buffer	Ramp Rate Control	✓	✓
	Oscillations		
Abnormal Event Response	Low Voltage Ride Through	✓	
	Voltage & Frequency Stability		✓
Bridge Power	Power Transition	✓	
	Black Start		✓
Grid Dispatchability	Peak Shaving	✓	
	Congestion Relief		

Smart Sequence to Power Up the AI Factory to Keep Energy Rates Low

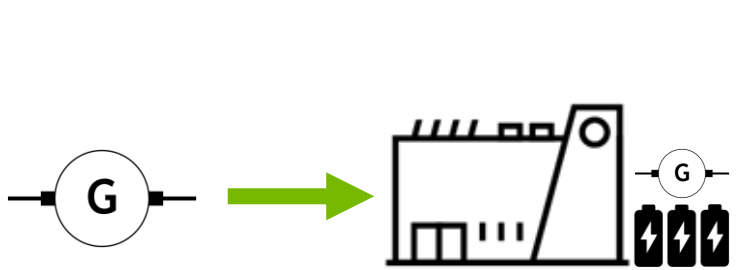


IF

1 - Grid Interconnection + ESS + Backup Gen

Grid First — *take the headroom and sell grid services*

Connect into existing grid slack; load spreads fixed cost and eases rates.

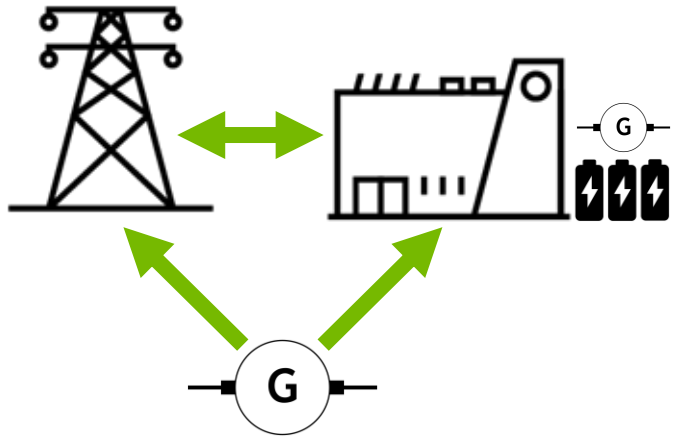


2a – Bridge-to-Grid + ESS + Backup Gen

Bridge to Grid — *power now, interconnect later*

Private generation when the grid can't meet the timeline, at no cost to ratepayers.

Then



3 – Flexible Grid Asset + BESS + Backup Gen

Flexible Grid Asset — *give capacity back*

Sell capacity and grid services back; monetize idle reserve, lower the cost to serve.

Key Takeaways & Call To Action

KEY TAKEAWAY

AI Factories are reliable and stable grid assets

Predictable load behavior and power quality

KEY TAKEAWAY

AI Factories are flexible grid assets

New data center designs features enable dispatchable load flexibility

CALL TO ACTION

Standardize interconnection requirements. Ratify large load classification

Align on a trusted global standard — built on transparent, ratified frameworks

Thank You!

